

Kokirala Vidhya Sagar

AI Engineer | Generative AI • RAG • Multi-Agent Systems • LLM Evaluation

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SUMMARY

AI Engineer with 4+ years of experience building production Generative AI systems (RAG chatbots, multi-agent workflows, LLM evaluation pipelines and AI-powered APIs) with Python, FastAPI, Azure OpenAI and LangGraph. Currently building AI products for LMS platforms at Red Nucleus; previously built IQE, an AI test automation platform, and an 8-agent QA framework at Atos Syntel that automated up to 80% of manual validation tasks. Microsoft Certified Azure AI Engineer Associate.

TECHNICAL SKILLS

AI Frameworks & Tools: LangGraph, LangChain, CrewAI, Microsoft Agent Framework, Letta, n8n, Hugging Face

AI Techniques: RAG, Multi-Agent Systems, Human-in-the-Loop, Prompt Engineering, Fine-tuning (LoRA, QLoRA)

LLMs & Platforms: Azure OpenAI, OpenAI, Claude, AWS Bedrock, Gemini, Llama, Qwen, Gemma

Evaluation & Observability: LLM Evaluation, RAGAS, DeepEval, Langfuse

Languages & Frameworks: Python, SQL, FastAPI, Django, Flask, JavaScript, HTML, CSS

Data & Infrastructure: PostgreSQL, pgvector, Redis, Docker, Git, CI/CD, Linux, Microsoft Azure, AWS

EXPERIENCE

AI Engineer | Red Nucleus

March 2026 – Present

Build production AI products for Learning Management System (LMS) platforms and internal teams using Python, FastAPI and Azure OpenAI.

- **RAG chatbot:** Built a retrieval-augmented generation (RAG) chatbot on FastAPI and Azure OpenAI that answers user questions from source documents, used across multiple client LMS platforms.
- **LLM evaluation:** Generated synthetic test datasets with RAGAS and automated chatbot response evaluation with DeepEval, enabling repeatable quality checks before each release.
- **AI answer evaluator:** Developed an LLM-based evaluator that automatically assesses learner answers in LMS modules, used by multiple clients.
- **Agentic personas:** Built a product on the Letta framework that maintains persistent, stateful persona context for interactive LMS simulations across multiple scenarios.
- **Text-to-SQL assistant:** Built a natural-language query chatbot in the LMS admin portal that converts plain-English questions into database queries and exports results as shareable files, giving managers self-service access to data.
- **Session survey tool:** Built an internal survey tool that collects attendee questions during company sessions and uses LLMs to categorize and summarize them.
- **Observability:** Implemented Langfuse tracing across all AI products for end-to-end visibility into prompts, retrieval steps and model responses.

Senior Consultant — AI Engineering | Atos Syntel

August 2022 – March 2026

Built Generative AI products for enterprise quality engineering (QE) teams, deployed across multiple customer environments.

- **IQE – Intelligent Quality Engineer (formerly QE Assist):** Built an AI-powered, node-based testing platform where QA teams design workflows on a drag-and-drop canvas to turn Jira requirements into reviewed test scenarios and test cases, reducing manual effort by 70%.
- Developed the platform across releases from Django and Flask/FastAPI to a fully async Python FastAPI application with 148 REST endpoints, PostgreSQL 16 and pgvector.
- Built a custom DAG execution engine with topological ordering and cycle detection, parallel execution of independent steps with asyncio, human-in-the-loop review pauses, sub-workflows and live step status over WebSocket.
- Developed 60+ pluggable node types, including AI test agents for scenario generation, test case generation, ambiguity detection and impact analysis, with LangGraph generate–validate–regenerate loops.

- Integrated Azure OpenAI, OpenAI, Anthropic Claude, Google Gemini, AWS Bedrock and Vertex AI through LangChain, with per-call token, latency and cost tracking.
- Built an 18-node RAG pipeline (document parsing, chunking, text-embedding-3-large embeddings, pgvector retrieval, reranking and query rewriting) to ground test generation in team documentation.
- Integrated with Jira, ALM, Rally, Azure Boards, Zephyr and Confluence for requirement sync, defect linking and pushing approved test cases.
- Deployed and fine-tuned small language models (Qwen, Llama, Gemma) using LoRA and QLoRA on customer-secured VMs.
- Shipped the platform as a multi-stage Docker image to Azure Container Apps, Web App for Containers and VMs through CI/CD, secured with JWT and Keycloak SSO, owner-scoped data access, Azure Key Vault and Microsoft Presidio PII masking.
- **QE Agents (multi-agent framework):** Designed a LangGraph multi-agent architecture using round-robin and role-based patterns, automating up to 80% of manual validation tasks across QA and DevOps workflows.
- Built 8 specialized agents for test case generation, requirement ambiguity detection, test data mining, defect validation, API validation, code vulnerability remediation, test coverage gap analysis and pull request review.
- Implemented agent memory and state persistence with Redis and vector stores, and integrated agent workflows into Django and FastAPI platforms.

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102)
- Microsoft Certified: Azure AI Fundamentals (AI-900); Microsoft Certified: Azure Fundamentals (AZ-900)
- AWS Certified AI Practitioner; AWS Certified Cloud Practitioner

EDUCATION

Master of Computer Applications (MCA) | University of Madras (Distance Education)

Bachelor of Science (BSc), Computer Science | Andhra Pradesh Residential Degree College

AWARDS

- **Employee of the Quarter**, Atos Syntel: recognized for performance and contributions to AI solutions.
- **Innovation Award**, Atos Syntel: for an AI solution that improved client satisfaction.
- **Spot Recognition (2x)**, Atos Syntel: for test script automation and for high-quality AI solution delivery.
- **Customer Demo Recognition**, Atos Syntel: for AI product demos that drove customer adoption.